

17

CHAPTER

Futures Options



The options we have considered so far provide the holder with the right to buy or sell a certain asset by a certain date for a certain price. They are sometimes termed *options on spot* or *spot options* because, when the options are exercised, the sale or purchase of the asset at the agreed-on price takes place immediately. In this chapter we move on to consider *options on futures*, also known as *futures options*. In these contracts, exercise of the option gives the holder a position in a futures contract.

The Commodity Futures Trading Commission in the US authorized the trading of options on futures on an experimental basis in 1982. Permanent trading was approved in 1987, and since then the popularity of the contract with investors has grown very fast.

In this chapter we consider how futures options work and the differences between these options and spot options. We examine how futures options can be priced using either binomial trees or formulas similar to those produced by Black, Scholes, and Merton for stock options. We also explore the relative pricing of futures options and spot options.

17.1 NATURE OF FUTURES OPTIONS

A futures option is the right, but not the obligation, to enter into a futures contract at a certain futures price by a certain date. Specifically, a call futures option is the right to enter into a long futures contract at a certain price; a put futures option is the right to enter into a short futures contract at a certain price. Futures options are generally American; that is, they can be exercised any time during the life of the contract.

If a call futures option is exercised, the holder acquires a long position in the underlying futures contract plus a cash amount equal to the most recent settlement futures price minus the strike price. If a put futures option is exercised, the holder acquires a short position in the underlying futures contract plus a cash amount equal to the strike price minus the most recent settlement futures price. As the following examples show, the effective payoff from a call futures option is $\max(F_T - K, 0)$ and the effective payoff from a put futures option is $\max(K - F_T, 0)$, where F_T is the futures price at the time of exercise and K is the strike price.

Example 17.1

Suppose it is August 15 and an investor has one September futures call option contract on copper with a strike price of 240 cents per pound. One futures contract

is on 25,000 pounds of copper. Suppose that the futures price of copper for delivery in September is currently 251 cents, and at the close of trading on August 14 (the last settlement) it was 250 cents. If the option is exercised, the investor receives a cash amount of

$$25,000 \times (250 - 240) \text{ cents} = \$2,500$$

plus a long position in a futures contract to buy 25,000 pounds of copper in September. If desired, the position in the futures contract can be closed out immediately. This would leave the investor with the \$2,500 cash payoff plus an amount

$$25,000 \times (251 - 250) \text{ cents} = \$250$$

reflecting the change in the futures price since the last settlement. The total payoff from exercising the option on August 15 is \$2,750, which equals $25,000(F - K)$, where F is the futures price at the time of exercise and K is the strike price.

Example 17.2

An investor has one December futures put option on corn with a strike price of 400 cents per bushel. One futures contract is on 5,000 bushels of corn. Suppose that the current futures price of corn for delivery in December is 380, and the most recent settlement price is 379 cents. If the option is exercised, the investor receives a cash amount of

$$5,000 \times (400 - 379) \text{ cents} = \$1,050$$

plus a short position in a futures contract to sell 5,000 bushels of corn in December. If desired, the position in the futures contract can be closed out. This would leave the investor with the \$1,050 cash payoff minus an amount

$$5,000 \times (380 - 379) \text{ cents} = \$50$$

reflecting the change in the futures price since the last settlement. The net payoff from exercise is \$1,000, which equals $5,000(K - F)$, where F is the futures price at the time of exercise and K is the strike price.

Expiration Months

Futures options are referred to by the delivery month of the underlying futures contract—not by the expiration month of the option. As mentioned earlier, most futures options are American. The expiration date of a futures option contract is usually on, or a few days before, the earliest delivery date of the underlying futures contract. (For example, the CME Group Treasury bond futures option expires on the latest Friday that precedes by at least five business days the end of the month before the futures delivery month.) An exception is the CME Group mid-curve Eurodollar contract where the futures contract expires either one or two years after the options contract.

Popular contracts trading in the United States are those on corn, soybeans, cotton, sugar-world, crude oil, natural gas, gold, Treasury bonds, Treasury notes, five-year Treasury notes, 30-day federal funds, Eurodollars, one-year and two-year mid-curve Eurodollars, Euribor, Eurobunds, and the S&P 500.

Options on Interest Rate Futures

The most actively traded interest rate options offered by exchanges in the United States are those on Treasury bond futures, Treasury note futures, and Eurodollar futures.

A Treasury bond futures option, which is traded by the CME Group, is an option to enter a Treasury bond futures contract. As mentioned in Chapter 6, one Treasury bond futures contract is for the delivery of \$100,000 of Treasury bonds. The price of a Treasury bond futures option is quoted as a percentage of the face value of the underlying Treasury bonds to the nearest sixty-fourth of 1%.

An option on Eurodollar futures, which is traded by the CME Group, is an option to enter into a Eurodollar futures contract. As explained in Chapter 6, when the Eurodollar futures quote changes by 1 basis point, or 0.01%, there is a gain or loss on a Eurodollar futures contract of \$25. Similarly, in the pricing of options on Eurodollar futures, 1 basis point represents \$25.

Interest rate futures option contracts work in the same way as the other futures options contracts discussed in this chapter. For example, in addition to the cash payoff, the holder of a call option obtains a long position in the futures contract when the option is exercised and the option writer obtains a corresponding short position. The total payoff from the call, including the value of the futures position, is $\max(F - K, 0)$, where F is the futures price at the time of exercise and K is the strike price.

Interest rate futures prices increase when bond prices increase (i.e., when interest rates fall). They decrease when bond prices decrease (i.e., when interest rates rise). An investor who thinks that short-term interest rates will rise can speculate by buying put options on Eurodollar futures, whereas an investor who thinks the rates will fall can speculate by buying call options on Eurodollar futures. An investor who thinks that long-term interest rates will rise can speculate by buying put options on Treasury note futures or Treasury bond futures, whereas an investor who thinks the rates will fall can speculate by buying call options on these instruments.

Example 17.3

It is February and the futures price for the June Eurodollar contract is 93.82 (corresponding to a 3-month Eurodollar interest rate of 6.18% per annum). The price of a call option on the contract with a strike price of 94.00 is quoted as 0.1, or 10 basis points. This option could be attractive to an investor who feels that interest rates are likely to come down. Suppose that short-term interest rates do drop by about 100 basis points and the investor exercises the call when the Eurodollar futures price is 94.78 (corresponding to a 3-month Eurodollar interest rate of 5.22% per annum). The payoff is $25 \times (94.78 - 94.00) \times 100 = \$1,950$. The cost of the contract is $10 \times 25 = \$250$. The investor's profit is therefore \$1,700.

Example 17.4

It is August and the futures price for the December Treasury bond contract is 96-09 (or $96\frac{9}{32} = 96.28125$). The yield on long-term government bonds is about 6.4% per annum. An investor who feels that this yield will fall by December might choose to buy December calls with a strike price of 98. Assume that the price of these calls is 1-04 (or $1\frac{4}{64} = 1.0625\%$ of the principal). If long-term rates

fall to 6% per annum and the Treasury bond futures price rises to 100-00, the investor will make a net profit per \$100 of bond futures of

$$100.00 - 98.00 - 1.0625 = 0.9375$$

Since one option contract is for the purchase or sale of instruments with a face value of \$100,000, the investor's profit is \$937.50 per option contract bought.

17.2 REASONS FOR THE POPULARITY OF FUTURES OPTIONS

It is natural to ask why people choose to trade options on futures rather than options on the underlying asset. The main reason appears to be that a futures contract is, in many circumstances, more liquid and easier to trade than the underlying asset. Furthermore, a futures price is known immediately from trading on the futures exchange, whereas the spot price of the underlying asset may not be so readily available.

Consider Treasury bonds. The market for Treasury bond futures is much more active than the market for any particular Treasury bond. Also, a Treasury bond futures price is known immediately from exchange trading. By contrast, the current market price of a bond can be obtained only by contacting one or more dealers. It is not surprising that investors would rather take delivery of a Treasury bond futures contract than Treasury bonds.

Futures on commodities are also often easier to trade than the commodities themselves. For example, it is much easier and more convenient to make or take delivery of a live-cattle futures contract than it is to make or take delivery of the cattle themselves.

An important point about a futures option is that exercising it does not usually lead to delivery of the underlying asset, as in most circumstances the underlying futures contract is closed out prior to delivery. Futures options are therefore normally eventually settled in cash. This is appealing to many investors, particularly those with limited capital who may find it difficult to come up with the funds to buy the underlying asset when an option on spot is exercised. Another advantage sometimes cited for futures options is that futures and futures options are traded side by side in the same exchange. This facilitates hedging, arbitrage, and speculation. It also tends to make the markets more efficient. A final point is that futures options entail lower transactions costs than spot options in many situations.

17.3 EUROPEAN SPOT AND FUTURES OPTIONS

The payoff from a European call option with strike price K on the spot price of an asset is

$$\max(S_T - K, 0)$$

where S_T is the spot price at the option's maturity. The payoff from a European call option with the same strike price on the futures price of the asset is

$$\max(F_T - K, 0)$$

where F_T is the futures price at the option's maturity. If the futures contract matures at

the same time as the option, then $F_T = S_T$ and the two options are equivalent. Similarly, a European futures put option is worth the same as its spot put option counterpart when the futures contract matures at the same time as the option.

Most of the futures options that trade are American-style. However, as we shall see, it is useful to study European futures options because the results that are obtained can be used to value the corresponding European spot options.

17.4 PUT-CALL PARITY

In Chapter 10, we derived a put-call parity relationship for European stock options. We now consider a similar argument to derive a put-call parity relationship for European futures options. Consider European call and put futures options, both with strike price K and time to expiration T . We can form two portfolios:

Portfolio A: a European call futures option plus an amount of cash equal to Ke^{-rT}

Portfolio B: a European put futures option plus a long futures contract plus an amount of cash equal to F_0e^{-rT} , where F_0 is the futures price

In portfolio A, the cash can be invested at the risk-free rate, r , and grows to K at time T . Let F_T be the futures price at maturity of the option. If $F_T > K$, the call option in portfolio A is exercised and portfolio A is worth F_T . If $F_T \leq K$, the call is not exercised and portfolio A is worth K . The value of portfolio A at time T is therefore

$$\max(F_T, K)$$

In portfolio B, the cash can be invested at the risk-free rate to grow to F_0 at time T . The put option provides a payoff of $\max(K - F_T, 0)$. The futures contract provides a payoff of $F_T - F_0$.¹ The value of portfolio B at time T is therefore

$$F_0 + (F_T - F_0) + \max(K - F_T, 0) = \max(F_T, K)$$

Because the two portfolios have the same value at time T and European options cannot be exercised early, it follows that they are worth the same today. The value of portfolio A today is

$$c + Ke^{-rT}$$

where c is the price of the call futures option. The daily settlement process ensures that the futures contract in portfolio B is worth zero today. Portfolio B is therefore worth

$$p + F_0e^{-rT}$$

where p is the price of the put futures option. Hence

$$c + Ke^{-rT} = p + F_0e^{-rT} \quad (17.1)$$

The difference between this put-call parity relationship and the one for a non-dividend-paying stock in equation (10.6) is that the stock price, S_0 , is replaced by the discounted futures price, F_0e^{-rT} .

¹ This analysis assumes that a futures contract is like a forward contract and settled at the end of its life rather than on a day-to-day basis.

As shown in Section 17.3, when the underlying futures contract matures at the same time as the option, European futures and spot options are the same. Equation (17.1) therefore gives a relationship between the price of a call option on the spot price, the price of a put option on the spot price, and the futures price when both options mature at the same time as the futures contract.

Example 17.5

Suppose that the price of a European call option on spot silver for delivery in six months is \$0.56 per ounce when the exercise price is \$8.50. Assume that the silver futures price for delivery in six months is currently \$8.00, and the risk-free interest rate for an investment that matures in six months is 10% per annum. From a rearrangement of equation (17.1), the price of a European put option on spot silver with the same maturity and exercise date as the call option is

$$0.56 + 8.50e^{-0.1 \times 6/12} - 8.00e^{-0.1 \times 6/12} = 1.04$$

For American futures options, the put-call relationship is (see Problem 17.19)

$$F_0e^{-rT} - K < C - P < F_0 - Ke^{-rT} \quad (17.2)$$

17.5 BOUNDS FOR FUTURES OPTIONS

The put-call parity relationship in equation (17.1) provides bounds for European call and put options. Because the price of a put, p , cannot be negative, it follows from equation (17.1) that

$$c + Ke^{-rT} \geq F_0e^{-rT}$$

or

$$c \geq (F_0 - K)e^{-rT} \quad (17.3)$$

Similarly, because the price of a call option cannot be negative, it follows from equation (17.1) that

$$Ke^{-rT} \leq F_0e^{-rT} + p$$

or

$$p \geq (K - F_0)e^{-rT} \quad (17.4)$$

These bounds are similar to the ones derived for European stock options in Chapter 10. The prices of European call and put options are very close to their lower bounds when the options are deep in the money. To see why this is so, we return to the put-call parity relationship in equation (17.1). When a call option is deep in the money, the corresponding put option is deep out of the money. This means that p is very close to zero. The difference between c and its lower bound equals p , so that the price of the call option must be very close to its lower bound. A similar argument applies to put options.

Because American futures options can be exercised at any time, we must have

$$C \geq F_0 - K$$

and

$$P \geq K - F_0$$

Thus, assuming interest rates are positive, the lower bound for an American option

price is always higher than the lower bound for the corresponding European option price. There is always some chance that an American futures option will be exercised early.

17.6 VALUATION OF FUTURES OPTIONS USING BINOMIAL TREES

This section examines, more formally than in Chapter 12, how binomial trees can be used to price futures options. A key difference between futures options and stock options is that there are no up-front costs when a futures contract is entered into.

Suppose that the current futures price is 30 and that it will move either up to 33 or down to 28 over the next month. We consider a one-month call option on the futures with a strike price of 29 and ignore daily settlement. The situation is as indicated in Figure 17.1. If the futures price proves to be 33, the payoff from the option is 4 and the value of the futures contract is 3. If the futures price proves to be 28, the payoff from the option is zero and the value of the futures contract is -2 .²

To set up a riskless hedge, we consider a portfolio consisting of a short position in one options contract and a long position in Δ futures contracts. If the futures price moves up to 33, the value of the portfolio is $3\Delta - 4$; if it moves down to 28, the value of the portfolio is -2Δ . The portfolio is riskless when these are the same, that is, when

$$3\Delta - 4 = -2\Delta$$

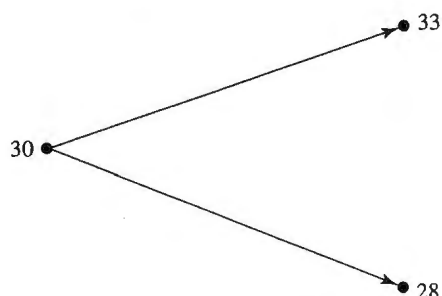
or $\Delta = 0.8$.

For this value of Δ , we know the portfolio will be worth $3 \times 0.8 - 4 = -1.6$ in one month. Assume a risk-free interest rate of 6%. The value of the portfolio today must be

$$-1.6e^{-0.06 \times 1/12} = -1.592$$

The portfolio consists of one short option and Δ futures contracts. Because the value of the futures contract today is zero, the value of the option today must be 1.592.

Figure 17.1 Futures price movements in numerical example.



² There is an approximation here in that the gain or loss on the futures contract is not realized at time T . It is realized day by day between time 0 and time T . However, as the length of the time step in a binomial tree becomes shorter, the approximation becomes better.

A Generalization

We can generalize this analysis by considering a futures price that starts at F_0 and is anticipated to rise to F_0u or move down to F_0d over the time period T . We consider an option maturing at time T and suppose that its payoff is f_u if the futures price moves up and f_d if it moves down. The situation is summarized in Figure 17.2.

The riskless portfolio in this case consists of a short position in one option combined with a long position in Δ futures contracts, where

$$\Delta = \frac{f_u - f_d}{F_0u - F_0d}$$

The value of the portfolio at time T is then always

$$(F_0u - F_0)\Delta - f_u$$

Denoting the risk-free interest rate by r , we obtain the value of the portfolio today as

$$[(F_0u - F_0)\Delta - f_u]e^{-rT}$$

Another expression for the present value of the portfolio is $-f$, where f is the value of the option today. It follows that

$$-f = [(F_0u - F_0)\Delta - f_u]e^{-rT}$$

Substituting for Δ and simplifying reduces this equation to

$$f = e^{-rT}[pf_u + (1 - p)f_d] \quad (17.5)$$

where

$$p = \frac{1 - d}{u - d} \quad (17.6)$$

This agrees with the result in Section 12.9. Equation (17.6) gives the risk-neutral probability of an up movement.

In the numerical example considered previously (see Figure 17.1), $u = 1.1$, $d = 0.9333$, $r = 0.06$, $T = 1/12$, $f_u = 4$, and $f_d = 0$. From equation (17.6),

$$p = \frac{1 - 0.9333}{1.1 - 0.9333} = 0.4$$

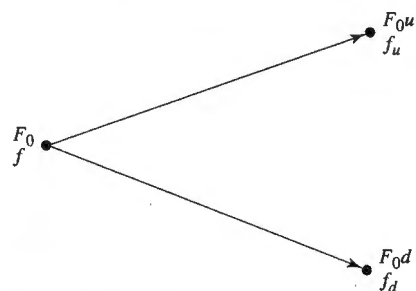


Figure 17.2 Futures price and option price in a general situation.

and, from equation (17.5),

$$f = e^{-0.06 \times 1/12} [0.4 \times 4 + 0.6 \times 0] = 1.592$$

This result agrees with the answer obtained for this example earlier.

Multistep Trees

Multistep binomial trees are used to value American-style futures options in much the same way that they are used to value options on stocks. This is explained in Section 12.9. The parameter u defining up movements in the futures price is $e^{\sigma\sqrt{\Delta t}}$, where σ is the volatility of the futures price and Δt is the length of one time step. The probability of an up movement in the future price is that in equation (17.6):

$$p = \frac{1 - d}{u - d}$$

Example 12.3 illustrates the use of multistep binomial trees for valuing a futures option. Example 20.3 in Chapter 20 provides a further illustration.

17.7 DRIFT OF A FUTURES PRICE IN A RISK-NEUTRAL WORLD

There is a general result that allows us to use the analysis in Section 16.3 for futures options. This result is that in a risk-neutral world a futures price behaves in the same way as a stock paying a dividend yield at the domestic risk-free interest rate r .

One clue that this might be so is given by noting that the equation for the probability p in a binomial tree for a futures price is the same as that for a stock paying a dividend yield equal to q when $q = r$ (compare equation (17.6) with equations (16.15) and (16.16)). Another clue is that the put-call parity relationship for futures options prices is the same as that for options on a stock paying a dividend yield at rate q when the stock price is replaced by the futures price and $q = r$ (compare equations (17.1) and (16.3)).

To prove the result formally, we calculate the drift of a futures price in a risk-neutral world. We define F_t as the futures price at time t and suppose the settlement dates to be at times $0, \Delta t, 2\Delta t, \dots$. If we enter into a long futures contract at time 0, its value is zero. At time Δt , it provides a payoff of $F_{\Delta t} - F_0$. If r is the very-short-term (Δt -period) interest rate at time 0, risk-neutral valuation gives the value of the contract at time 0 as

$$e^{-r\Delta t} \hat{E}[F_{\Delta t} - F_0]$$

where $\hat{E}$ denotes expectations in a risk-neutral world. We must therefore have

$$e^{-r\Delta t} \hat{E}(F_{\Delta t} - F_0) = 0$$

showing that

$$\hat{E}(F_{\Delta t}) = F_0$$

Similarly, $\hat{E}(F_{2\Delta t}) = F_{\Delta t}$, $\hat{E}(F_{3\Delta t}) = F_{2\Delta t}$, and so on. Putting many results like this together, we see that

$$\hat{E}(F_T) = F_0$$

for any time T .

The drift of the futures price in a risk-neutral world is therefore zero. From equation (16.7), the futures price behaves like a stock providing a dividend yield q equal to r . This result is a very general one. It is true for all futures prices and does not depend on any assumptions about interest rates, volatilities, etc.³

The usual assumption made for the process followed by a futures price F in the risk-neutral world is

$$dF = \sigma F dz \quad (17.7)$$

where σ is a constant.

Differential Equation

For another way of seeing that a futures price behaves like a stock paying a dividend yield at rate q , we can derive the differential equation satisfied by a derivative dependent on a futures price in the same way as we derived the differential equation for a derivative dependent on a non-dividend-paying stock in Section 14.6. This is⁴

$$\frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial F^2} \sigma^2 F^2 = rf \quad (17.8)$$

It has the same form as equation (16.6) with q set equal to r . This confirms that, for the purpose of valuing derivatives, a futures price can be treated in the same way as a stock providing a dividend yield at rate r .

17.8 BLACK'S MODEL FOR VALUING FUTURES OPTIONS

European futures options can be valued by extending the results we have produced. Fischer Black was the first to show this in a paper published in 1976.⁵ Assuming that the futures price follows the (lognormal) process in equation (17.7), the European call price c and the European put price p for a futures option are given by equations (16.4) and (16.5) with S_0 replaced by F_0 and $q = r$:

$$c = e^{-rT} [F_0 N(d_1) - KN(d_2)] \quad (17.9)$$

$$p = e^{-rT} [KN(-d_2) - F_0 N(-d_1)] \quad (17.10)$$

where

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma \sqrt{T}}$$

$$d_2 = \frac{\ln(F_0/K) - \sigma^2 T/2}{\sigma \sqrt{T}} = d_1 - \sigma \sqrt{T}$$

and σ is the volatility of the futures price. When the cost of carry and the convenience

³ As we will discover in Chapter 27, a more precise statement of the result is: "A futures price has zero drift in the traditional risk-neutral world where the numeraire is the money market account." A zero-drift stochastic process is known as a martingale. A forward price is a martingale in a different risk-neutral world. This is one where the numeraire is a zero-coupon bond maturing at time T .

⁴ See Technical Note 7 at www.rotman.utoronto.ca/~hull/TechnicalNotes for a proof of this.

⁵ See F. Black, "The Pricing of Commodity Contracts," *Journal of Financial Economics*, 3 (March 1976): 167-79.

yield arc functions only of time, it can be shown that the volatility of the futures price is the same as the volatility of the underlying asset.

Example 17.6

Consider a European put futures option on crude oil. The time to the option's maturity is 4 months, the current futures price is \$20, the exercise price is \$20, the risk-free interest rate is 9% per annum, and the volatility of the futures price is 25% per annum. In this case, $F_0 = 20$, $K = 20$, $r = 0.09$, $T = 4/12$, $\sigma = 0.25$, and $\ln(F_0/K) = 0$, so that

$$d_1 = \frac{\sigma\sqrt{T}}{2} = 0.07216$$

$$d_2 = -\frac{\sigma\sqrt{T}}{2} = -0.07216$$

$$N(-d_1) = 0.4712, \quad N(-d_2) = 0.5288$$

and the put price p is given by

$$p = e^{-0.09 \times 4/12} (20 \times 0.5288 - 20 \times 0.4712) = 1.12$$

or \$1.12.

Using Black's Model Instead of Black-Scholes-Merton

The results in Section 17.3 show that European futures options and European spot options are equivalent when the option contract matures at the same time as the futures contract. Equations (17.9) and (17.10) therefore provide a way of calculating the value of European options on the spot price of an asset.

Example 17.7

Consider a six-month European call option on the spot price of gold, that is, an option to buy one ounce of gold in the spot market in six months. The strike price is \$1,200, the six-month futures price of gold is \$1,240, the risk-free rate of interest is 5% per annum, and the volatility of the futures price is 20%. The option is the same as a six-month European option on the six-month futures price. The value of the option is therefore given by equation (17.9) as

$$e^{-0.05 \times 0.5} [1,240N(d_1) - 1,200N(d_2)]$$

where

$$d_1 = \frac{\ln(1,240/1,200) + 0.2^2 \times 0.5/2}{0.2 \times \sqrt{0.5}} = 0.3026$$

$$d_2 = \frac{\ln(1,240/1,200) - 0.2^2 \times 0.5/2}{0.2 \times \sqrt{0.5}} = 0.1611$$

It is \$88.37.

Traders like to use Black's model rather than Black-Scholes-Merton to value European spot options. It has a fairly general applicability. The underlying asset can be a consumption or investment asset and it can provide income to the holder. The variable F_0 in equations (17.9) and (17.10) is set equal to either the futures or the forward price of the underlying asset for a contract maturing at the same time as the option.

Equations (16.13) and (16.14) show Black's model being used to value European options on the spot value of a currency. Equations (16.8) and (16.9) show Black's model being used to value European options on the spot value of an index. The big advantage of Black's model is that it avoids the need to estimate the income (or convenience yield) on the underlying asset. The futures or forward price that is used in the model incorporate the market's estimate of this income.

When considering stock indices in Section 16.4, we explained that put-call parity is used to imply the forward prices for maturities for which there are actively traded options. Interpolation is then used to estimate forward prices for other maturities. The same approach can be used for a wide range of other underlying assets.

17.9 AMERICAN FUTURES OPTIONS vs. AMERICAN SPOT OPTIONS

Traded futures options are in practice usually American. Assuming that the risk-free rate of interest, r , is positive, there is always some chance that it will be optimal to exercise an American futures option early. American futures options are therefore worth more than their European counterparts.

It is not generally true that an American futures option is worth the same as the corresponding American spot option when the futures and options contracts have the same maturity.⁶ Suppose, for example, that there is a normal market with futures prices consistently higher than spot prices prior to maturity. An American call futures option must be worth more than the corresponding American spot call option. The reason is that in some situations the futures option will be exercised early, in which case it will provide a greater profit to the holder. Similarly, an American put futures option must be worth less than the corresponding American spot put option. If there is an inverted market with futures prices consistently lower than spot prices, the reverse must be true. American call futures options are worth less than the corresponding American spot call option, whereas American put futures options are worth more than the corresponding American spot put option.

The differences just described between American futures options and American spot options hold true when the futures contract expires later than the options contract as well as when the two expire at the same time. In fact, the later the futures contract expires the greater the differences tend to be.

17.10 FUTURES-STYLE OPTIONS

Some exchanges trade what are termed *futures-style options*. These are futures contracts on the payoff from an option. Normally a trader who buys (sells) an option, whether on the spot price of an asset or on the futures price of an asset, pays (receives) cash up front. By contrast, traders who buy or sell a futures-style option post margin in the same way that they do on a regular futures contract (see Chapter 2). The contract is settled daily as with any other futures contract and the final settlement price is the payoff from the option. Just as a futures contract is a bet on what the future price of an

⁶ The spot option "corresponding" to a futures option is defined here as one with the same strike price and the same expiration date.

asset will be, a futures-style option is a bet on what the payoff from an option will be.⁷ If interest rates are constant, the futures price in a futures-style option is the same as the forward price in a forward contract on the option payoff. This shows that the futures price for a futures-style option is the price that would be paid for the option if payment were made in arrears. It is therefore the value of a regular option compounded forward at the risk-free rate.

Black's model in equations (17.9) and (17.10) gives the price of a regular European option on an asset in terms of the futures (or forward) price F_0 for a contract maturing at the same time as the option. The futures price in a call futures-style option is therefore

$$F_0 N(d_1) - KN(d_2)$$

and the futures price in a put futures-style option is

$$KN(-d_2) - F_0 N(-d_1)$$

where d_1 and d_2 are as defined in equations (17.9) and (17.10). These formulas do not depend on the level of interest rates. They are correct for a futures-style option on a futures contract and a futures-style option on the spot value of an asset. In the first case, F_0 is the current futures price for the contract underlying the option; in the second case, it is the current futures price for a futures contract on the underlying asset maturing at the same time as the option.

The put-call parity relationship for a futures-style options is

$$p + F_0 = c + K$$

An American futures-style option can be exercised early, in which case there is an immediate final settlement at the option's intrinsic value. As it turns out, it is never optimal to exercise an American futures-style options on a futures contract early because the futures price of the option is always greater than the intrinsic value. This type of American futures-style option can therefore be treated as though it were the corresponding European futures-style option.

SUMMARY

Futures options require delivery of the underlying futures contract on exercise. When a call is exercised, the holder acquires a long futures position plus a cash amount equal to the excess of the futures price over the strike price. Similarly, when a put is exercised the holder acquires a short position plus a cash amount equal to the excess of the strike price over the futures price. The futures contract that is delivered usually expires slightly later than the option.

A futures price behaves in the same way as a stock that provides a dividend yield equal to the risk-free rate, r . This means that the results produced in Chapter 16 for options on a stock paying a dividend yield apply to futures options if we replace the stock price by the futures price and set the dividend yield equal to the risk-free interest

⁷ For a more detailed discussion of futures-style options, see D. Lieu, "Option Pricing with Futures-Style Margining," *Journal of Futures Markets*, 10, 4 (1990), 327-38. For pricing when interest rates are stochastic, see R.-R. Chen and L. Scott, "Pricing Interest Rate Futures Options with Futures-Style Margining," *Journal of Futures Markets*, 13, 1 (1993) 15-22.

rate. Pricing formulas for European futures options were first produced by Fischer Black in 1976. They assume that the futures price is lognormally distributed at the option's expiration.

If the expiration dates for the option and futures contracts are the same, a European futures option is worth exactly the same as the corresponding European spot option. This result is often used to value European spot options. The result is not true for American options. If the futures market is normal, an American call futures is worth more than the corresponding American spot call option, while an American put futures is worth less than the corresponding American spot put option. If the futures market is inverted, the reverse is true.

FURTHER READING

Black, F. "The Pricing of Commodity Contracts," *Journal of Financial Economics*, 3 (1976): 167–79.

Hilliard, J. E., and J. Reis. "Valuation of Commodity Futures and Options under Stochastic Convenience Yields, Interest Rates, and Jump Diffusions in the Spot," *Journal of Financial and Quantitative Analysis*, 33, 1 (March 1998): 61–86.

Miltersen, K. R., and E. S. Schwartz. "Pricing of Options on Commodity Futures with Stochastic Term Structures of Convenience Yields and Interest Rates," *Journal of Financial and Quantitative Analysis*, 33, 1 (March 1998): 33–59.

Practice Questions (Answers in Solutions Manual)

- 17.1. Explain the difference between a call option on yen and a call option on yen futures.
- 17.2. Why are options on bond futures more actively traded than options on bonds?
- 17.3. "A futures price is like a stock paying a dividend yield." What is the dividend yield?
- 17.4. A futures price is currently 50. At the end of six months it will be either 56 or 46. The risk-free interest rate is 6% per annum. What is the value of a six-month European call option on the futures with a strike price of 50?
- 17.5. How does the put–call parity formula for a futures option differ from put–call parity for an option on a non-dividend-paying stock?
- 17.6. Consider an American futures call option where the futures contract and the option contract expire at the same time. Under what circumstances is the futures option worth more than the corresponding American option on the underlying asset?
- 17.7. Calculate the value of a five-month European put futures option when the futures price is \$19, the strike price is \$20, the risk-free interest rate is 12% per annum, and the volatility of the futures price is 20% per annum.
- 17.8. Suppose you buy a put option contract on October gold futures with a strike price of \$1,200 per ounce. Each contract is for the delivery of 100 ounces. What happens if you exercise when the October futures price is \$1,180?
- 17.9. Suppose you sell a call option contract on April live cattle futures with a strike price of 90 cents per pound. Each contract is for the delivery of 40,000 pounds. What happens if the contract is exercised when the futures price is 95 cents?

- 17.10. Consider a two-month call futures option with a strike price of 40 when the risk-free interest rate is 10% per annum. The current futures price is 47. What is a lower bound for the value of the futures option if it is (a) European and (b) American?
- 17.11. Consider a four-month put futures option with a strike price of 50 when the risk-free interest rate is 10% per annum. The current futures price is 47. What is a lower bound for the value of the futures option if it is (a) European and (b) American?
- 17.12. A futures price is currently 60 and its volatility is 30%. The risk-free interest rate is 8% per annum. Use a two-step binomial tree to calculate the value of a six-month European call option on the futures with a strike price of 60. If the call were American, would it ever be worth exercising it early?
- 17.13. In Problem 17.12, what does the binomial tree give for the value of a six-month European put option on futures with a strike price of 60? If the put were American, would it ever be worth exercising it early? Verify that the call prices calculated in Problem 17.12 and the put prices calculated here satisfy put-call parity relationships.
- 17.14. A futures price is currently 25, its volatility is 30% per annum, and the risk-free interest rate is 10% per annum. What is the value of a nine-month European call on the futures with a strike price of 26?
- 17.15. A futures price is currently 70, its volatility is 20% per annum, and the risk-free interest rate is 6% per annum. What is the value of a five-month European put on the futures with a strike price of 65?
- 17.16. Suppose that a one-year futures price is currently 35. A one-year European call option and a one-year European put option on the futures with a strike price of 34 are both priced at 2 in the market. The risk-free interest rate is 10% per annum. Identify an arbitrage opportunity.
- 17.17. "The price of an at-the-money European call futures option always equals the price of a similar at-the-money European put futures option." Explain why this statement is true.
- 17.18. Suppose that a futures price is currently 30. The risk-free interest rate is 5% per annum. A three-month American call futures option with a strike price of 28 is worth 4. Calculate bounds for the price of a three-month American put futures option with a strike price of 28.
- 17.19. Show that, if C is the price of an American call option on a futures contract when the strike price is K and the maturity is T , and P is the price of an American put on the same futures contract with the same strike price and exercise date, then

$$F_0 e^{-rT} - K < C - P < F_0 - K e^{-rT}$$

where F_0 is the futures price and r is the risk-free rate. Assume that $r > 0$ and that there is no difference between forward and futures contracts. (*Hint:* Use an analogous approach to that indicated for Problem 16.12.)

- 17.20. Calculate the price of a three-month European call option on the spot value of silver. The three-month futures price is \$12, the strike price is \$13, the risk-free rate is 4% and the volatility of the price of silver is 25%.
- 17.21. A corporation knows that in three months it will have \$5 million to invest for 90 days at LIBOR minus 50 basis points and wishes to ensure that the rate obtained will be at least 6.5%. What position in exchange-traded options should it take to hedge?

Further Questions

- 17.22. A futures price is currently 40. It is known that at the end of three months the price will be either 35 or 45. What is the value of a three-month European call option on the futures with a strike price of 42 if the risk-free interest rate is 7% per annum?
- 17.23. It is February 4. July call options on corn futures with strike prices of 260, 270, 280, 290, and 300 cost 26.75, 21.25, 17.25, 14.00, and 11.375, respectively. July put options with these strike prices cost 8.50, 13.50, 19.00, 25.625, and 32.625, respectively. The options mature on June 19, the current July corn futures price is 278.25, and the risk-free interest rate is 1.1%. Calculate implied volatilities for the options using DerivaGem. Comment on the results you get.
- 17.24. Calculate the implied volatility of soybean futures prices from the following information concerning a European put on soybean futures:

Current futures price	525
Exercise price	525
Risk-free rate	6% per annum
Time to maturity	5 months
Put price	20

- 17.25. Calculate the price of a six-month European put option on the spot value of the S&P 500. The six-month forward price of the index is 1,400, the strike price is 1,450, the risk-free rate is 5%, and the volatility of the index is 15%.
- 17.26. The strike price of a futures option is 550 cents, the risk-free interest rate is 3%, the volatility of the futures price is 20%, and the time to maturity of the option is 9 months. The futures price is 500 cents.
- What is the price of the option if it is a European call?
 - What is the price of the option if it is a European put?
 - Verify that put-call parity holds.
 - What is the futures price for a futures-style option if it is a call?
 - What is the futures price for a futures-style option if it is a put?